

Chapter 3

The relativistic particle and gauge symmetry

3.1 The totally antisymmetric tensor

The totally antisymmetric tensor is defined by

$$\epsilon^{\alpha\beta\gamma\delta} = \begin{cases} 1 & \text{for } 0123[?] \\ -1 & \text{for odd permutations} \\ 0 & \text{otherwise} \end{cases} \quad (3.1)$$

Transforming it as a $(4, 0)$ tensor gives

$$\begin{aligned} \epsilon^{\alpha'\beta'\gamma'\delta'} &= \Lambda^{\alpha'}_{\mu} \Lambda^{\beta'}_{\nu} \Lambda^{\gamma'}_{\rho} \Lambda^{\delta'}_{\sigma} \epsilon^{\mu\nu\rho\sigma} \\ &= (\det \hat{\Lambda}) \epsilon^{\alpha'\beta'\gamma'\delta'} = \epsilon^{\alpha'\beta'\gamma'\delta'} \end{aligned} \quad (3.2)$$

using $\det \hat{\Lambda} = +1$: like the metric, $\epsilon^{\alpha\beta\gamma\delta}$ is a tensor that does not change under Lorentz transformations. NOTE: $\hat{\Lambda}$ is not a tensor!

3.2 Point particle action

The action of a free relativistic point particle is

$$S = -m c \int d\tau \quad , \quad \text{a } (0, 0) \text{ invariant} \quad (3.3)$$

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Writing out the proper time interval,

$$\begin{aligned} S &= -m c \int \sqrt{dt^2 - \frac{1}{c^2} (dx^2 + dy^2 + dz^2)} \\ &= -m c \int dt \sqrt{1 - \frac{\dot{x}^i \dot{x}^i}{c^2}} \end{aligned} \quad (3.4)$$

where $\dot{} \equiv d/dt$, so that the Lagrangian is

$$L = -m c \sqrt{1 - \frac{\dot{x}^i \dot{x}^i}{c^2}} \quad (3.5)$$

The equations of motion follow from

$$\frac{\partial L}{\partial \dot{x}^j} = \frac{m \dot{x}^j}{\sqrt{1 - \dot{x}^i \dot{x}^i / c^2}} = \gamma m \dot{x}^j = p^j \quad , \quad \frac{\partial L}{\partial x^j} = 0 \quad (3.6)$$

which gives

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}^j} \right) = \frac{\partial L}{\partial x^j} \Rightarrow \frac{d}{dt} (\vec{p}) = 0 \quad (3.7)$$

– the relativistic free particle.

We want this particle to interact with some background force field. Lorentz symmetry requires the simplest possibility to be

$$S \rightarrow S = -m c \int d\tau + \frac{q}{c} \int dx^\mu A_\mu \quad (3.8)$$

where q is the charge ($q \rightarrow 0$ cancels the force), the factor q/c is convention, the integral is over the path in spacetime, and $dx^\mu A_\mu$ is a $(0,0)$ invariant. Hence, a $(0,1)$ tensor field in the background is needed

$$A_\mu(x) \equiv \text{potential} \quad (3.9)$$

The new term can be written as

$$\begin{aligned} \frac{q}{c} \int dx^\mu A_\mu &= \frac{q}{c} \int dt \frac{dx^\mu}{dt} A_\mu = \frac{q}{c} \int dt (c A_0 + v^i A_i) \\ &= \int dt (q A_0 + q \dot{x}^i A_i) \end{aligned} \quad (3.10)$$

It is more convenient to parameterize the path by proper time. The action (3.8) becomes

$$\begin{aligned} S &= -m \int \sqrt{-\eta_{\mu\nu} dx^\mu dx^\nu} + \frac{q}{c} \int A_\mu dx^\mu \\ &= -m \int d\tau \sqrt{-\eta_{\mu\nu} \dot{x}^\mu \dot{x}^\nu} + \frac{q}{c} \int d\tau A_\mu \dot{x}^\mu \end{aligned} \quad (3.11)$$

where now $\dot{} = d/d\tau$, with Lagrangian

$$L = -m \sqrt{-\eta_{\mu\nu} \dot{x}^\mu \dot{x}^\nu} + \frac{q}{c} A_\mu(x) \dot{x}^\mu \quad (3.12)$$

The equations of motion follow from

$$\frac{\partial L}{\partial \dot{x}^\mu} = p_\mu + \frac{q}{c} A_\mu \Rightarrow \frac{d}{d\tau} \left(\frac{\partial L}{\partial \dot{x}^\mu} \right) = \dot{p}_\mu + \frac{q}{c} \dot{x}^\nu \partial_\nu A_\mu \quad (3.13)$$

and

$$\frac{\partial L}{\partial x^\mu} = \frac{q}{c} \dot{x}^\nu \partial_\mu A_\nu \quad (3.14)$$

so that

$$\begin{aligned} \dot{p}_\mu &= \frac{q}{c} \dot{x}^\nu \partial_\mu A_\nu - \frac{q}{c} \dot{x}^\nu \partial_\nu A_\mu \\ &= \frac{q}{c} \dot{x}^\nu (\partial_\mu A_\nu - \partial_\nu A_\mu) \equiv \frac{q}{c} \dot{x}^\nu F_{\mu\nu} \end{aligned} \quad (3.15)$$

where $F_{\mu\nu}$ is the $(0, 2)$ field strength.

3.3 Gauge symmetry

There is a new symmetry in this system: gauge symmetry. The shift

$$A_\mu \rightarrow A_\mu + \partial_\mu \Lambda \quad \text{for any bounded } \Lambda \quad (3.16)$$

does not change the physics. Indeed, in the action

$$\int dx^\mu A_\mu \rightarrow \int dx^\mu A_\mu + \int dx^\mu \partial_\mu \Lambda \quad (3.17)$$

and the last term is a total derivative, vanishing at infinity. And the equations of motion (3.15) depend on

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \quad , \quad F_{\mu\nu} = -F_{\nu\mu} \quad \text{antisymmetric} \quad (3.18)$$

which is invariant

$$F_{\mu\nu} \rightarrow \partial_\mu A_\nu - \partial_\nu A_\mu + \partial_\mu \partial_\nu \Lambda - \partial_\nu \partial_\mu \Lambda = F_{\mu\nu} \quad (3.19)$$